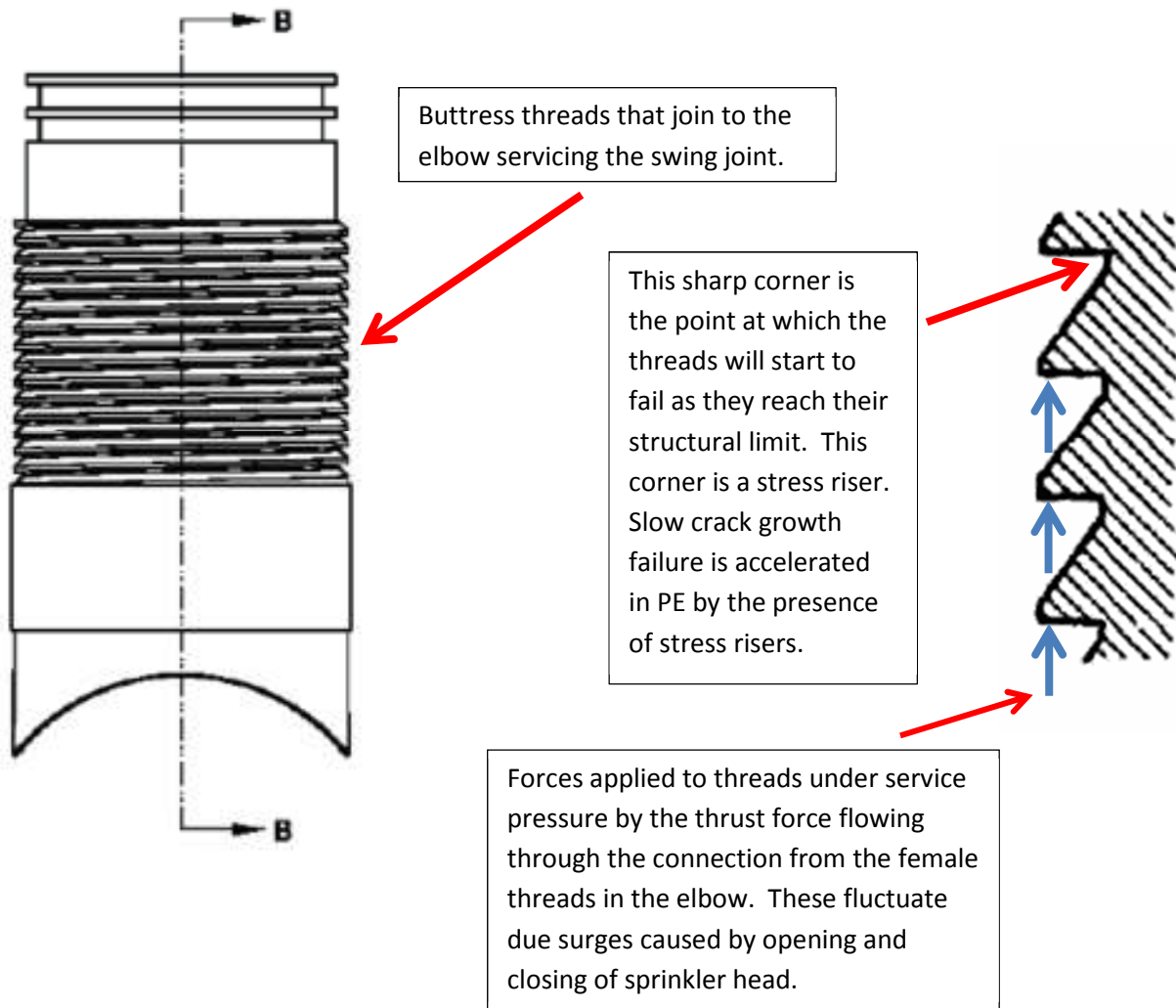


CMF/Aquafuse Fusible Service Saddle - 2

In addition to the side wall fusion connection to the pipe the saddle employs a buttress thread. CMF will argue that this buttress thread and the long thread length make their threads ok proven by their limited testing and recent experience. ASTM F1498 advises: "Some pressure rated pipe materials (for example PE) are not recommended for threaded applications." Buttress threads are used in the gas and oil industry for steel casing pipe. Although buttress threads have increased holding capacity for steel, they have a sharp corner that is a stress riser that will reduce the Slow Crack Growth life in PE.



In addition to the thrust force borne by the threads, the saddle bears the internal system pressure hoop stress. Both forces fluctuate with surges adding a fatigue component of loading. The presence of the threads will reduce the life of the component as applied in a system. The only way to determine the capability of the saddle as used is to perform appropriate long term testing. The limiting factor will be Slow Crack Growth performance of the saddle fusion and the threads.